

SDR Hunter — Quick Start

1. Install

```
# Python deps (works without SDR hardware using the built-in mock device)
pip install -r requirements.txt

# Full SDR driver stack (SoapySDR + BladeRF/RTL-SDR/HackRF/LimeSDR/Pluto/USRP/SDRplay/Airspy/NooElec)
./install_drivers.sh
```

Arch / CachyOS

Arch-based distros (including **CachyOS**) differ in two ways: they enforce **PEP 668** (a system-wide pip install is blocked, so you must use a venv), and CachyOS defaults to the **fish** shell (the usual activate script won't work). `./install_drivers.sh` handles this automatically, but the manual steps are:

```
# 1. System packages (note the Arch names: libuhd, python-pyqt6-webengine, portaudio)
sudo pacman -S --needed soapysdr soapyrtlsdr soapyhackrf
soapybladerf soapyuhd libuhd \
    rtl-sdr hackrf libusb portaudio python-pyqt6 python-pyqt6-
webengine

# 2. Create a venv WITH --system-site-packages (so it can see the
pacman SoapySDR/PyQt6 bindings)
python -m venv --system-site-packages .venv

# 3. Activate it — the command depends on your shell:
source .venv/bin/activate          # bash / zsh
source .venv/bin/activate.fish     # fish (CachyOS default)
# ...or skip activation and call .venv/bin/python directly.

# 4. Install the Python deps
pip install -r requirements.txt
```

Seeing 'case' builtin not inside of switch block? You're in **fish** but sourced the bash activate script — use `activate.fish` instead. Do **not** use `pip install --break-system-packages`; use the venv above.

LimeSDR / PlutoSDR / SDRplay / Airspy Soapy modules are AUR-only, e.g. `paru -S soapylms7 soapysdrplay3 soapyairspy limesuite` (optional).

2. Run

```
./launch.sh # desktop GUI (same as: python3
main.py --gui)
python3 main.py --web # web interface only (http://
localhost:8000)
python3 main.py --both # GUI + web server
python3 main.py --list-devices
```

No SDR attached? The app automatically falls back to a synthetic **mock device**, or force it with:

```
SDRHUNTER_FORCE MOCK=1 ./launch.sh
```

3. First scan

1. Pick a device in the **Device** toolbar drop-down.
2. Set **RX0** center frequency (e.g. 100.0 MHz) and sample rate.
3. Press ► **Start Scan** (or Space).
4. Detected signals appear in the right-hand list; double-click one to focus **RX1**.

4. Tabs

- **Main** — dual-RX spectrum + waterfall + signal list
- **Drone Tracking** — map + Remote ID / OpenDroneID contacts (F3)
- **Audio Decoder** — AM/FM/SSB demod + classification
- **Weather Satellite** — NOAA APT / METEOR-M2 LRPT
- **Spectrum Hunting** — band-hopping hunt mode (F2)
- **Signal Database** — browse/search 90+ known signals

5. Keyboard shortcuts

Key	Action
F1	Standard view
F2	Spectrum Hunting view
F3	Drone Tracking view
F4	Dual Signal Analysis
Space	Start/Stop scanning

Key	Action
R	Start/Stop recording
Ctrl+T	Tune RX1
Ctrl+B	Bookmark current frequency
Ctrl+L	Load baseline
Esc	Cancel current operation

6. Remote / web access

Enable the  **Web Server** toolbar button (or --web) and browse to `http://<host>:8000` from another machine on the network.